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Oefeningenreeks 2F nr. 3.2

Te bewijzen: $\coth^2 x - \text{csch}^2 x = 1$

Bewijs:

$$\coth^2 x - \text{csch}^2 x = \left(\frac{\cosh x}{\sinh x}\right)^2 - \left(\frac{1}{\sinh x}\right)^2$$
$$= \frac{\cosh^2 x}{\sinh^2 x} - \frac{1}{\sinh^2 x}$$
$$= \frac{\cosh^2 x - 1}{\sinh^2 x}$$

$\cosh^2 x - 1 = \sinh^2 x$

$$\frac{\sinh^2 x}{\sinh^2 x} = 1$$

References